

Cybersecurity Mini Project Ideas (Basic, Medium, Hard)

Basic Level Projects

1. Password Strength Checker
2. Simple Port Scanner
3. File Integrity Checker
4. Basic Encryption Tool
5. Secure Notes App
6. Log-In Brute Force Prevention Demo
7. URL Phishing Detector (Basic Version)
8. Basic Firewall Rule Simulation
9. Simple Keylogger (Isolated VM Only)
10. Network Device Scanner

Medium Level Projects

1. Build a Vulnerable Web App (Local Only)
2. Packet Sniffer using Python & Scapy
3. Basic Intrusion Detection System (IDS)
4. Secure File Transfer System
5. Simple Web Application Firewall (WAF) Simulator
6. JWT Security Analyzer
7. Vulnerability Scanner (Local Only)
8. Android App Pen-Test Lab (Local APK)
9. Phishing Awareness Simulation
10. Secure Coding Checker

Hard Level Projects

1. Build Your Own Vulnerability Scanner (Advanced)
2. Create a Malware Analysis Sandbox
3. Develop a SIEM Dashboard
4. Implement Your Own Ransomware (Educational VM Only)
5. Build a Blockchain-Based Authentication System
6. Create a Custom Network Intrusion Prevention System (IPS)
7. Mobile App Security Scanner
8. Threat Intelligence Automation Tool
9. Reverse Engineer a Binary (Legal, Public Sample)
10. Build Your Own CTF Platform